

GRAND PRIX

Hackathon Problem Statements

Theme: Artificial Intelligence in Racing Strategy & Decision-Making | Powered by Hugging Face

Problem Statements

PROBLEM STATEMENT 1

The Silent Co-Driver: Reading Driver Stress from Radio Calls

The Challenge

During a race, the team talks to the driver over radio. Sometimes the driver sounds tired, frustrated, or stressed, but the engineers are too busy watching numbers to notice the tone of voice. Important warning signs get missed simply because no one has time to listen carefully while also watching the data.

The Solution

- You can play or upload a radio audio clip.
- The system converts the speech to text (so it's readable).
- The system also studies the tone of voice and shows if the driver seems calm, stressed, or tired.
- This is shown alongside basic lap-time information, so the team can see if stress is matching up with slower laps.

Frontend: A clean screen where you upload or play the audio, and see the text, the detected mood, and a simple chart of lap times.

Backend: Where the audio is processed and the AI models do the thinking.

Input: Audio clips of driver radio messages, lap time data.

Output: Text version of what was said, a mood or stress label (like Calm, Stressed, or Tired), and a simple visual showing if mood is affecting lap performance.

PROBLEM STATEMENT 2

Weather Whiplash: Live Track Condition Detector

The Challenge

Track conditions (dry, wet, drying) can change very fast, faster than weather reports can keep up. Teams need to know right now if the track is becoming safer or riskier, so they can decide when to change tires.

The Solution

- You feed in images or short video frames from a camera (trackside or onboard).
- The AI looks at each image and tells you if the track looks Dry, Damp, Wet, or Drying.
- Over time, it shows a simple trend: is the track getting better or worse?
- Based on this, it gives a basic suggestion like "Consider tire change soon."

Frontend: A simple screen showing the uploaded image, the predicted condition, and a trend line or graph over time.

Backend: Where the images are processed and the AI decides the condition.

Input: Photos or video frames of the track, basic weather info (optional).

Output: A label for each image (Dry, Damp, Wet, or Drying), a simple trend graph, and a suggestion message (e.g., "Track drying: tire change window approaching").

PROBLEM STATEMENT 3

Crowd Flow Optimiser: Simulating and Rerouting Crowds in Real Time

The Challenge

Large venues and events, stadiums, railway stations, festivals, often see people bunch up at entry gates, food counters, or exits without warning. When that happens without anyone noticing in time, it can turn into a real safety risk. Right now, there is no easy way to spot these pile-ups before they become dangerous.

The Solution

- You feed in the layout of the venue: entry gates, walkways, concession points, and emergency exits.
- The system simulates how crowds move through this layout under different crowd sizes and event schedules.
- It automatically points out where and when bottlenecks are likely to form.
- It suggests real-time rerouting, so people are guided away from crowded areas before things get risky.

Frontend: A screen showing the venue layout, with crowded zones highlighted and suggested alternate routes marked out clearly.

Backend: Where the crowd simulation runs, and where bottlenecks and rerouting suggestions are calculated.

Input: Venue layout (gates, walkways, concession points, emergency exits), expected crowd size, and event schedule.

Output: A map showing bottleneck zones and recommended rerouting paths, updated as crowd conditions change.

Applications: Stadium and concert venue management, railway station design, and crowd safety at large Indian gatherings such as the Kumbh Mela, IPL matches, and airport terminals.

Tips to Stand Out

Your idea can end up being like what other teams are building. That is completely normal and is not a reason to worry. What separates a winning team from the rest is usually not a completely unique idea, it is how well the idea is executed. Here are a few things you can focus on to give your team the edge:

- **Frontend:** Even if the idea is common, a clean, simple, and easy to understand frontend can make your solution feel far more polished and usable than others.
- **Execution:** A solution that works end to end, with the frontend and backend properly connected, will stand out more than a solution with a bigger idea that only partly works.
- **Research:** Spend some time understanding the real problem before building. Teams that show they understood the actual pain point (through their explanation, their data choices, or their demo) tend to leave a stronger impression.
- **Iteration:** Do not settle for your first working version. Going back, testing it, finding what feels clunky or wrong, and improving it even slightly can make a noticeable difference by the end.
- **Clarity of presentation:** Being able to explain what your solution does, why it matters, and how it works, in simple language, often matters as much as the build itself.

So even if you see another team with a similar idea, do not panic or change direction at the last minute. Focus on building it well, testing it properly, and presenting it clearly. That is usually what gets a team through to the next round.

General Rules: Mandatory for All Teams

- The solution must be a combination of both a frontend (what the user sees and interacts with) and a backend (where the data is processed and the AI does its work). A backend-only or notebook-only submission will not be accepted.
- Teams should aim for a solution that is balanced in difficulty, not something that can be solved by simply calling one ready-made tool, but also not something that requires building an AI model completely from scratch.
- Every single team member must have their own individual Hugging Face account. The solution must use something (this could be a model, a dataset, a Space, or any suitable tool) available on the Hugging Face Hub. There is no restriction on which specific Hugging Face model or tool must be used.